The hook tracer coefficient: closed form $g_k(q) = (q+1)^{\binom{k}{2}}$

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Abstract

For every hook $\lambda = (k, 1^{n-k})$ in the rank-zero regime $n \geq 2k$, the matrix coefficient

$$g_k(q) \;:=\; \langle v_{T_*^{(\lambda)}},\, B^{(\lambda)}_{j_0(\lambda)}\, v_{T^{(\lambda)}_{\text{first}}} \rangle$$

between the two tracer SYTs of `2026-05-13-hook-dim-equals-one.tex` equals

$$g_k(q) \;=\; (q+1)^{\binom{k}{2}},$$

in the asymmetric Murphy normalization of the seminormal basis (the convention $b' = 1$ for every 2-block off-diagonal). This upgrades the "$g_k(q) \neq 0$" theorem of the hook-dim-1 paper to an exact closed form, settling its Corollary 5.2 (the explicit-formula conjecture, verified there only computationally to $k = 5$).

The proof refines Lemmas A and B of the hook-dim-1 paper. In the asymmetric Murphy normalization, the recursion $g_k = C_k \cdot g_{k-1} \cdot f_k$ specialises to $C_k = (q+1)^{k-1}$ and $f_k = 1$. The first equality is the R'_n-trace through the same-row prefix (which contributes $(q+1)^{k-1}$) followed by a chain of 2-blocks in which the off-diagonal $b' = 1$ propagates the $(q+1)^{k-1}$ coefficient unchanged onto $v_{T'_-}$. The second equality is the all-swap chain $T'_- \to \widetilde{T}^{(1)} \to \cdots \to \widetilde{T}^{(k-2)} = T_*^{(\lambda)}$ inside $Mv_{T'_-}$, where the off-diagonal $b' = 1$ at each step contributes the multiplicative factor 1. Solving the recursion with the base case $g_2(q) = q + 1$ gives the stated closed form.

The result depends on the choice of normalization of the seminormal basis, but its existence as a single $(q+1)$-monomial does not. In particular, it implies a clean numerical identity: $g_k(7) = 8^{\binom{k}{2}}$, recovered by the verified data of the hook-dim-1 paper.

1 Setup

1.1 Seminormal normalization

We work in the seminormal basis of V_λ used throughout the May-2026 stream of papers (the asymmetric Murphy convention from `verify_seminormal.py`). On each T_i-2-block $\operatorname{span}(v_T, v_{s_iT})$ at axial distance $d \neq \pm 1$, the matrix of T_i is

$$T_i\,(v_T,\, v_{s_iT}) \;=\; (v_T,\, v_{s_iT}) \begin{pmatrix} a_d & b_d b'_d \\ 1 & a'_d \end{pmatrix}, \tag{1}$$

with diagonal entries

$$a_d \;=\; \frac{(q-1)q^d}{q^d - 1}, \qquad a'_d \;=\; \frac{q-1}{1-q^d} \;=\; -\,a_d/q^d,$$

off-diagonal $b'_d = 1$, and the remaining off-diagonal $b_d b'_d = a_d a'_d + q = q - a_d^2/q^d$. The eigenvalues of T_i on this 2-block are q and -1.

In this normalization, the action of (T_i+1) on the 2-block is:

$$\begin{aligned}(T_i+1)v_T &= (a_d+1)\,v_T + 1\cdot v_{s_iT},\\ (T_i+1)v_{s_iT} &= b_d b'_d\cdot v_T + (a'_d+1)\,v_{s_iT}.\end{aligned}$$

A direct computation in q gives the identity

$$q - a'_d \;=\; a_d + 1, \tag{2}$$

so the $+q$-eigenvector of T_i on the 2-block, normalized to have coefficient 1 on v_{s_iT}, is $(a_d+1)\,v_T + v_{s_iT}$.

1.2 Tracer SYTs and the conjectural formula

For $\lambda = (k, 1^{n-k})$, $k \geq 2$, $n \geq 2k$, we use the hook-S notation: an SYT of λ is determined by its $(k-1)$-element subset $S \subset \{2, \ldots, n\}$ filling row 1 after the entry 1 at $(1,1)$. The two tracer SYTs of `2026-05-13-hook-dim-equals-one.tex` are

$$T^{(\lambda)}_{\text{first}}:\; S = \{2,3,\ldots,k\}, \qquad T^{(\lambda)}_{*}:\; S = \{n-k+2,\ldots,n\}.$$

The tracer coefficient is

$$g_k(q) \;=\; \langle v_{T^{(\lambda)}_*},\; B^{(\lambda)}_{j_0(\lambda)}\, v_{T^{(\lambda)}_{\text{first}}}\rangle,$$

where $B^{(\lambda)}_{j_0} = R'_{j_0}\,R'_{j_0+1}\cdots R'_n$ with $j_0(\lambda) = n-k+2$ and $R'_k = (T_{k-1}+1)(T_{k-2}+1)\cdots(T_1+1)$.

Theorem 1 (Main)**.** *For every $k \geq 2$ and every $n \geq 2k$,*

$$g_k(q) \;=\; (q+1)^{\binom{k}{2}}.$$

In particular g_k is a polynomial in $\mathbb{Z}_{\geq 0}[q]$ independent of n inside the rank-zero regime.

1.3 The recursion

The hook-dim-1 paper's Lemma A and Lemma B set up a recursion

$$g_k(q) \;=\; C_k\cdot g_{k-1}(q)\cdot f_k(q), \tag{3}$$

where:

- C_k is the scalar in Lemma A: $R'_n\,v_{T^{(\lambda)}_{\text{first}}} = C_k\cdot\Phi_*(v_{T^{(\mu_1)}_{\text{first}}})$, where $\mu_1 = (k-1, 1^{n-k-1})$ and Φ_* is the embedding from Definition 2 of the hook-dim-1 paper;
- f_k is the scalar in Lemma B: $\langle v_{T^{(\lambda)}_*}, M\,\Phi_*(v_{T^{(\mu_1)}_*})\rangle = f_k$, with $M = (T_\ell+1)(T_{\ell+1}+1)\cdots(T_{n-2}+1)$ and $\ell = n-k+1 = j_0(\lambda)-1$.

We sharpen the qualitative statements "$C_k \neq 0$, $f_k \neq 0$" to the quantitative

$$C_k = (q+1)^{k-1}, \qquad f_k = 1$$

in the normalization fixed above.

2 Refined Lemma A: $C_k = (q+1)^{k-1}$

We normalize Φ_* by

$$\Phi_*(v_{T'}) \;=\; (a_{-(n-1)}+1)\, v_{T'_+} + 1\cdot v_{T'_-}\,, \tag{4}$$

the $+q$-eigenvector of T_{n-1} on the corner pair $\{(1,k),(n-k+1,1)\}$ (axial distance $-(n-1)$), scaled to have coefficient 1 on $v_{T'_-}$. This is consistent with the convention of the hook-dim-1 paper (Definition 2) once a b'-normalization is fixed.

Lemma 2 (Refined Lemma A)**.** *For every $k \ge 2$ and $n \ge 2k$,*

$$R'_n\, v_{T^{(\lambda)}_{\text{first}}} \;=\; (q+1)^{k-1}\cdot \Phi_*(v_{T^{(\mu_1)}_{\text{first}}}).$$

In particular, $C_k = (q+1)^{k-1}$.

Proof. We trace $R'_n = (T_{n-1}{+}1)(T_{n-2}{+}1)\cdots(T_1{+}1)$ applied right-to-left to $v_{T^{(\lambda)}_{\text{first}}}$, identifying the seminormal support and coefficients at each step.

Same-row prefix (steps 1 through $k-1$). For each $j \in [1,k-1]$, entries j and $j+1$ both lie in row 1 of $T^{(\lambda)}_{\text{first}}$ at adjacent cells $(1,j)$ and $(1,j+1)$. Hence $(T_j+1)v_{T^{(\lambda)}_{\text{first}}} = (q+1)\, v_{T^{(\lambda)}_{\text{first}}}$. Cumulatively:

$$(T_{k-1}+1)\cdots(T_1+1)\, v_{T^{(\lambda)}_{\text{first}}} \;=\; (q+1)^{k-1}\, v_{T^{(\lambda)}_{\text{first}}}\,.$$

2-block phase (steps k through $n-1$). Define, for $m \ge 1$, the SYT $T^{(m)}$ of λ by the row-1 data

$$T^{(m)}:\; S = \{2,3,\dots,k-1,k+m\}.$$

Thus $T^{(0)} = T^{(\lambda)}_{\text{first}}$, and $T^{(m)}$ is obtained from $T^{(m-1)}$ by transposing the entries $k+m-1$ and $k+m$ (this is $s_{k+m-1}T^{(m-1)}$ on tableaux). The positions in $T^{(m)}$ are:

- Row 1: $(1,1)=1, (1,2)=2, \dots, (1,k-1)=k-1, (1,k)=k+m$.
- Column 1: $(2,1)=k, (3,1)=k+1, \dots, (m+1,1)=k+m-1, (m+2,1)=k+m+1, \dots, (n-k+1,1)=n$.

We claim by induction on j that for $j = k, k+1, \dots, n-1$,

$$(T_j+1)\cdots(T_1+1)\, v_{T^{(\lambda)}_{\text{first}}} \;=\; \alpha_j\, v_{T^{(j-k)}} + (q+1)^{k-1}\, v_{T^{(j-k+1)}}\,, \tag{5}$$

for some $\alpha_j \in \mathbb{Q}(q)$ whose precise form we record below.

Step $j=k$ (base of induction). Apply (T_k+1) to $(q+1)^{k-1}\, v_{T_{\text{first}}}$. In $T_{\text{first}} = T^{(0)}$, entry k is at $(1,k)$ (content $k-1$) and entry $k+1$ is at $(2,1)$ (content -1). Axial distance $d_k = -1-(k-1) = -k \ne \pm 1$, so T_k acts as a 2-block with partner $T^{(1)}$. Using (1):

$$(T_k+1)\, v_{T^{(0)}} \;=\; (a_{-k}+1)\, v_{T^{(0)}} + 1\cdot v_{T^{(1)}}\,,$$

hence

$$(T_k+1)\cdots(T_1+1)\, v_{T^{(\lambda)}_{\text{first}}} \;=\; (q+1)^{k-1}(a_{-k}+1)\, v_{T^{(0)}} + (q+1)^{k-1}\, v_{T^{(1)}}\,,$$

matching (5) with $\alpha_k = (q+1)^{k-1}(a_{-k}+1)$.

Inductive step $j \to j+1$, $j \in [k, n-2]$. Suppose (5) holds at step j. Apply $(T_{j+1}+1)$. By the position data of $T^{(m)}$ above:

- In $T^{(j-k)}$, entry $j+1$ is at $(j-k+2,1)$ and entry $j+2$ is at $(j-k+3,1)$ (both in column 1, adjacent). Axial distance -1: $(T_{j+1}+1)v_{T^{(j-k)}} = 0$.
- In $T^{(j-k+1)}$, entry $j+1$ is at $(1,k)$ (content $k-1$) and entry $j+2$ is at $(j-k+3,1)$ (content $-(j-k+2)$, from the column-1 description of $T^{(j-k+1)}$: $(2,1)=k,\ldots,(j-k+2,1)=j,(j-k+3,1)=j+2$, with the gap at the row-1 cell $(1,k)=j+1$). Axial distance $d_{j+1} = -(j-k+2)-(k-1) = -(j+1)$. Hence T_{j+1} acts as a 2-block with partner $T^{(j-k+2)}$:

$$(T_{j+1}+1)\,v_{T^{(j-k+1)}} \;=\; (a_{-(j+1)}+1)\,v_{T^{(j-k+1)}} + 1\cdot v_{T^{(j-k+2)}}.$$

Combining: the state at step $j+1$ equals

$$(q+1)^{k-1}(a_{-(j+1)}+1)\,v_{T^{(j-k+1)}} + (q+1)^{k-1}\,v_{T^{(j-k+2)}},$$

which matches (5) at $j+1$ with $\alpha_{j+1} = (q+1)^{k-1}(a_{-(j+1)}+1)$ and the same persistent coefficient $(q+1)^{k-1}$ on the new SYT. This completes the induction: $\alpha_j = (q+1)^{k-1}(a_{-j}+1)$ for every $j \in [k, n-1]$.

The final step $j = n-1$**.** At step $n-1$, we apply $(T_{n-1}+1)$ to the state from step $n-2$. By (5) at $j = n-2$, the input is supported on $\{T^{(n-k-2)}, T^{(n-k-1)}\}$. The state after applying $(T_{n-1}+1)$ is supported on $\{T^{(n-k-1)}, T^{(n-k)}\}$, with coefficients (per the inductive formula at $j = n-1$):

$$R'_n\, v_{T^{(\lambda)}_{\text{first}}} \;=\; (q+1)^{k-1}(a_{-(n-1)}+1)\,v_{T^{(n-k-1)}} + (q+1)^{k-1}\,v_{T^{(n-k)}}.$$

Identification with Φ_***.** The tableaux $T^{(n-k-1)}$ and $T^{(n-k)}$ are precisely T'_+ and T'_- for $T' = T^{(\mu_1)}_{\text{first}}$:

- $T^{(n-k-1)}$: $S = \{2,\ldots,k-1,n-1\}$, so entry $n-1$ at $(1,k)$ and entry n at $(n-k+1,1)$. This is T'_+.
- $T^{(n-k)}$: $S = \{2,\ldots,k-1,n\}$, so entry n at $(1,k)$ and entry $n-1$ at $(n-k+1,1)$. This is T'_-.

Therefore

$$R'_n\, v_{T^{(\lambda)}_{\text{first}}} \;=\; (q+1)^{k-1}\cdot\big[\,(a_{-(n-1)}+1)\,v_{T'_+} + 1\cdot v_{T'_-}\,\big] \;=\; (q+1)^{k-1}\cdot\Phi_*(v_{T^{(\mu_1)}_{\text{first}}}),$$

using the normalization (4) of Φ_*. □

3 Refined Lemma B: $f_k = 1$

Lemma 3 (Refined Lemma B)**.** *For every $k \geq 2$ and $n \geq 2k$, with the normalization of Φ_* in (4):*

$$\langle v_{T^{(\lambda)}_*},\; M\,\Phi_*(v_{T^{(\mu_1)}_*})\rangle \;=\; 1.$$

In particular, $f_k = 1$.

Proof. By Step 1 of Lemma B in the hook-dim-1 paper, only the $v_{T'_-}$ summand of $\Phi_*(v_{T^{(\mu_1)}_*}) = (a_{-(n-1)}+1)v_{T'_+} + v_{T'_-}$ contributes to the coefficient at $v_{T^{(\lambda)}_*}$: the factors of M do not involve T_{n-1}, so entry n is fixed under M; in T'_+ it is at $(n-k+1,1)$, in $T^{(\lambda)}_*$ it is at $(1,k)$. Hence

$$\langle v_{T^{(\lambda)}_*},\, M\,\Phi_*(v_{T^{(\mu_1)}_*})\rangle \;=\; 1\cdot\langle v_{T^{(\lambda)}_*},\, M\,v_{T'_-}\rangle.$$

The T'_- data. For $T' = T_*^{(\mu_1)}$, the tableau T'_- has row $1 = \{1, n-k+1, n-k+2, \ldots, n-2, n\}$ and column $1 = \{1, 2, 3, \ldots, n-k, n-1\}$. Explicitly,

$$(1,j) = n-k+j-1 \text{ for } j \in [2, k-1], \quad (1,k) = n,$$

$$(i,1) = i \text{ for } i \in [1, n-k], \quad (n-k+1, 1) = n-1.$$

The all-swap chain. Define the SYTs $\widetilde{T}^{(0)} = T'_-$ and inductively for $m = 1, 2, \ldots, k-2$:

$$\widetilde{T}^{(m)} \ := \ s_{n-1-m}\, \widetilde{T}^{(m-1)},$$

i.e., $\widetilde{T}^{(m)}$ is obtained from $\widetilde{T}^{(m-1)}$ by swapping the entries $n-1-m$ and $n-m$. Tracking positions:

Lemma 4 (Position tracking)**.** *For each $m \in \{0, 1, \ldots, k-2\}$, the tableau $\widetilde{T}^{(m)}$ has:*

- *Row 1: $(1,j) = n-k+j-1$ for $j \in [2, k-m-1]$, $(1, k-i) = n-i$ for $i \in [1, m]$, and $(1,k) = n$.*
- *Column 1: $(i,1) = i$ for $i \in [1, n-k]$, $(n-k+1, 1) = n-1-m$.*

In particular, $\widetilde{T}^{(k-2)}$ has row $1 = \{1, n-k+2, n-k+3, \ldots, n-1, n\}$ and column $1 = \{1, 2, \ldots, n-k+1\}$, which is precisely $T_*^{(\lambda)}$.

Proof of Lemma 4. By induction on m. Base $m = 0$: the description matches T'_- above. Inductive step: applying s_{n-1-m} to $\widetilde{T}^{(m-1)}$ swaps entries $n-1-m$ and $n-m$. In $\widetilde{T}^{(m-1)}$, entry $n-m$ sits at $(n-k+1, 1)$ (by the IH at $m-1$, since $n-1-(m-1) = n-m$), and entry $n-1-m$ sits at $(1, k-m)$ (the row-1 slot not yet "rotated up"). After the swap, $(1, k-m) = n-m$ and $(n-k+1, 1) = n-1-m$, matching the claim at m. At $m = k-2$ the row-1 entries are exactly $\{1, n-k+2, n-k+1+1, \ldots, n-1, n\} = \{1, n-k+2, n-k+3, \ldots, n-1, n\}$ (in order), and the column-1 bottom is $n-k+1$, which is $T_*^{(\lambda)}$. $\square$

M acts on the all-swap chain by off-diagonal $b' = 1$. Write $M = (T_\ell{+}1)(T_{\ell+1}{+}1)\cdots(T_{n-2}{+}1)$ applied right-to-left (i.e., $(T_{n-2}{+}1)$ first, $(T_\ell{+}1)$ last). Index steps of M by $m = 1, 2, \ldots, k-2$, with step m applying the Hecke generator T_{n-1-m}. By Lemma 4 applied to $\widetilde{T}^{(m-1)}$, the entries $n-1-m$ and $n-m$ sit at row 1 cell $(1, k-m)$ and column 1 bottom $(n-k+1, 1)$ respectively, with axial distance

$$d_m \ = \ c(n-m) - c(n-1-m) \ = \ -(n-k) - (k-m-1) \ = \ -(n-1-m).$$

Since $n \geq 2k$ and $m \leq k-2$, we have $n-1-m \geq 2$, so $d_m \leq -2$ — a non-degenerate 2-block. Hence

$$(T_{n-1-m} + 1)\, v_{\widetilde{T}^{(m-1)}} \ = \ (a_{d_m} + 1)\, v_{\widetilde{T}^{(m-1)}} + 1 \cdot v_{\widetilde{T}^{(m)}}. \tag{6}$$

Support and uniqueness of the all-swap branch. We claim that for each $m \in \{0, 1, \ldots, k-2\}$, the support of $(T_{n-1-m}{+}1)\cdots(T_{n-2}{+}1)\, v_{T'_-}$ is exactly $\{v_{\widetilde{T}^{(0)}}, v_{\widetilde{T}^{(1)}}, \ldots, v_{\widetilde{T}^{(m)}}\}$.

Indeed, by induction: at step m, the input support is $\{\widetilde{T}^{(0)}, \ldots, \widetilde{T}^{(m-1)}\}$. We split into cases by j.

Case $j = m-1$ (the newest SYT, on which T_{n-1-m} has been computed as a 2-block by (6)): the 2-block produces $\widetilde{T}^{(m)}$ as the partner with off-diagonal coefficient $b' = 1$ (the new SYT).

Case $j \leq m-2$ (older SYTs): we claim the action is same-row. Apply Lemma 4 to $\widetilde{T}^{(j)}$. Since $m \geq j+2$, both $n-1-m$ and $n-m$ are strictly less than the down-shifted entry $n-1-j$ at $(n-k+1, 1)$, and both are strictly less than the touched-range minimum $n-j$ (which is $\geq n-m+1$

here). Hence both entries lie in the untouched row-1 range $(1, j') = n-k+j'-1$ for $j' \in [2, k-j-1]$. Solving:

$$n - 1 - m = n - k + j' - 1 \;\Rightarrow\; j' = k - m,$$
$$n - m = n - k + j' - 1 \;\Rightarrow\; j' = k - m + 1.$$

Both $j' = k - m$ and $k - m + 1$ lie in $[2, k - j - 1]$ since $m \in [j + 2, k - 2]$. So in $\widetilde{T}^{(j)}$, entry $n - 1 - m$ is at $(1, k - m)$ and entry $n - m$ is at $(1, k - m + 1)$. These are adjacent row-1 cells, hence $(T_{n-1-m} + 1)\, v_{\widetilde{T}^{(j)}} = (q + 1)\, v_{\widetilde{T}^{(j)}}$. No new SYT introduced.

Hence after step m, the support is $\{\widetilde{T}^{(0)}, \ldots, \widetilde{T}^{(m)}\}$, as claimed. In particular, $\widetilde{T}^{(m)}$ is the *unique* new SYT, and its coefficient picks up only from the off-diagonal $b' = 1$ in (6) applied to $\widetilde{T}^{(m-1)}$.

Computing the coefficient at $T_*^{(\lambda)} = \widetilde{T}^{(k-2)}$. By the induction, $v_{\widetilde{T}^{(k-2)}}$ first appears at step $k-2$ with coefficient 1 (the off-diagonal b' of (6)) times the coefficient of $\widetilde{T}^{(k-3)}$ at step $k-3$, which is in turn 1 times the coefficient of $\widetilde{T}^{(k-4)}$ at step $k-4$, etc. Since each transition picks up exactly $b' = 1$ and the initial coefficient of $\widetilde{T}^{(0)} = T'_-$ is 1, we get

$$\langle v_{T_*^{(\lambda)}},\; M\, v_{T'_-} \rangle \;=\; \prod_{m=1}^{k-2} 1 \;=\; 1.$$

(For $k = 2$, M is the empty product, so $M v_{T'_-} = v_{T'_-}$ and $T'_- = T_*^{(\lambda)}$ directly, giving coefficient 1 as well.) □

4 Proof of Theorem 1

Proof of Theorem 1. Induction on k.

Base case $k = 2$. $\lambda = (2, 1^{n-2})$, $j_0(\lambda) = n$. $T_{\text{first}}^{(\lambda)}$ has $S = \{2\}$; $T_*^{(\lambda)}$ has $S = \{n\}$. Apply Lemma 2 with $k = 2$: $R'_n\, v_{T_{\text{first}}^{(\lambda)}} = (q + 1) \cdot \Phi_*(v_{T_{\text{first}}^{(\mu_1)}})$. But for $k = 2$, $\mu_1 = (1, 1^{n-3}) = (1^{n-2})$ is the $(n-2)$-sign representation; its unique SYT is $T_{\text{first}}^{(\mu_1)} = T_*^{(\mu_1)}$, and $\Phi_*(v_{T'}) = (a_{-(n-1)} + 1) v_{T'_+} + v_{T'_-}$ with $T'_- = T_*^{(\lambda)}$. Hence

$$g_2(q) \;=\; \langle v_{T_*^{(\lambda)}}, R'_n\, v_{T_{\text{first}}^{(\lambda)}} \rangle \;=\; (q + 1) \cdot 1 \;=\; q + 1.$$

This matches $(q + 1)^{\binom{2}{2}} = (q + 1)^1 = q + 1$.

Inductive step $k \geq 3$. Assume Theorem 1 for $k - 1$ (in its rank-zero regime $n - 2 \geq 2(k - 1)$, i.e., $n \geq 2k$): $g_{k-1}(q) = (q + 1)^{\binom{k-1}{2}}$.

By Lemma 2,

$$R'_n\, v_{T_{\text{first}}^{(\lambda)}} \;=\; (q + 1)^{k-1} \cdot \Phi_*(v_{T_{\text{first}}^{(\mu_1)}}).$$

By the Φ_*-commutation of the May-13 r1 paper (`2026-05-13-r1-inductive-reduction.tex`), the iteration $R'_{j_0(\lambda)} \cdots R'_{n-1}$ pulls through Φ_* into the outer factor M:

$$R'_{j_0(\lambda)} \cdots R'_{n-1}\, \Phi_*(v_{T_{\text{first}}^{(\mu_1)}}) \;=\; M\, \Phi_*\Big(R'^{(\mu_1)}_{j_0(\mu_1)} \cdots R'^{(\mu_1)}_{n-2}\, v_{T_{\text{first}}^{(\mu_1)}}\Big) \;=\; M\, \Phi_*(w_{\mu_1}),$$

where $w_{\mu_1} := B_{j_0(\mu_1)}^{(\mu_1)}\, v_{T_{\text{first}}^{(\mu_1)}} \in V_{\mu_1}$. Composing:

$$B_{j_0(\lambda)}^{(\lambda)}\, v_{T_{\text{first}}^{(\lambda)}} \;=\; (q + 1)^{k-1} \cdot M\, \Phi_*(w_{\mu_1}).$$

Expand $w_{\mu_1} = \sum_{T' \in \mathrm{SYT}(\mu_1)} c_{T'} \, v_{T'}$. By the inductive hypothesis, $c_{T_*^{(\mu_1)}} = g_{k-1}(q) = (q+1)^{\binom{k-1}{2}}$. Apply Φ_* linearly, then M, then extract the coefficient at $v_{T_*^{(\lambda)}}$:

$$g_k(q) \;=\; (q+1)^{k-1} \cdot \sum_{T' \in \mathrm{SYT}(\mu_1)} c_{T'} \, \langle v_{T_*^{(\lambda)}}, \, M \, \Phi_*(v_{T'}) \rangle.$$

By Lemma B of the hook-dim-1 paper (Step 2's combinatorial argument: M does not move entries $\le n-k$, and only $T' = T_*^{(\mu_1)}$ has its column-1 small entries match $T_*^{(\lambda)}$), the sum collapses to the $T' = T_*^{(\mu_1)}$ term. By Lemma 3, the surviving inner product equals 1. Therefore

$$g_k(q) \;=\; (q+1)^{k-1} \cdot c_{T_*^{(\mu_1)}} \cdot 1 \;=\; (q+1)^{k-1} \cdot (q+1)^{\binom{k-1}{2}} \;=\; (q+1)^{\binom{k-1}{2}+(k-1)} \;=\; (q+1)^{\binom{k}{2}},$$

using the Pascal-like identity $\binom{k-1}{2} + (k-1) = \binom{k}{2}$. $\square$

5 Computational verification

Scripts. All verification scripts use `verify_seminormal.py`'s asymmetric Murphy normalization ($b' = 1$).

- `~/projects/scratch/prove-2026-05-13-tracer-formula/verify_formula.py`: computes $g_k(q)$ symbolically and confirms $g_k(q) = (q+1)^{\binom{k}{2}}$ for all tested (k, n) with diff $= 0$.
- `verify_Ck_fk.py`: verifies Lemmas 2 and 3 directly, by computing the coefficient of $v_{T'_-}$ in $R'_n v_{T^{(\lambda)}_{\mathrm{first}}}$ (expected $(q+1)^{k-1}$) and the coefficient of $v_{T_*^{(\lambda)}}$ in $M v_{T'_-}$ (expected 1).

Verified cases (Theorem 1). $(k, n) \in \{(2,4), (2,5), (2,6), (3,6), (3,7), (3,8), (4,8), (4,9), (5,10)\}$. Exact match of $g_k(q) = (q+1)^{\binom{k}{2}}$ in every case.

Verified cases (Lemmas 2 and 3). $(k, n) \in \{(2,4), (2,5), (3,6), (3,7), (3,8), (4,8), (4,9), (5,10)\}$. $C_k = (q+1)^{k-1}$ and $f_k = 1$ in every case.

6 Honest status and consequences

Proved unconditionally (in the asymmetric Murphy normalization of the seminormal basis). $g_k(q) = (q+1)^{\binom{k}{2}}$ for every hook $\lambda = (k, 1^{n-k})$ with $k \ge 2$, $n \ge 2k$. This closes Corollary 5.2 of `2026-05-13-hook-dim-equals-one.tex`, which had verified the formula computationally up to $k = 5$ but left the proof open.

Normalization-dependence. The exact constant $(q+1)^{\binom{k}{2}}$ depends on the choice of seminormal off-diagonals. In the symmetric Murphy convention ($b = b' = \sqrt{b_d b'_d}$ with consistent sign), the formula picks up a product of $\sqrt{b_d b'_d}$ factors. However, the *shape* of the formula — a single monomial in $(q+1)$ with exponent $\binom{k}{2}$ — is structural. Any seminormal normalization yields $g_k(q) = (q+1)^{\binom{k}{2}} \cdot \prod(\text{Hoefsmit prefactors})$.

Why the answer is n-independent. The crucial structural fact is that:

(i) The same-row prefix in Lemma 2 contributes $(q+1)^{k-1}$, independent of n (the prefix has length $k-1$, the number of same-row pairs in row 1 of T_{first}).

(ii) The 2-block phase in Lemma 2 "slides" the new-SYT coefficient unchanged across $n-k$ steps via the off-diagonal $b'=1$, contributing no n-dependent factor.

(iii) The all-swap chain in Lemma 3 has length $k-2$, independent of n.

The n-dependent Hoefsmit prefactors (a_d+1) for $d=-k,-(k+1),\ldots,-(n-1)$ all appear in the orthogonal α-coefficients α_j but NOT in the persistent $\beta_j=(q+1)^{k-1}$, because the asymmetric Murphy off-diagonal $b'_d=1$ is itself d-independent. The structural "rigidity" of the formula is a feature of this normalization choice.

Beyond hooks. For non-hook shapes with multiple corners, $\dim B$ can exceed 1, and the tracer-pair framework requires generalization. See the May-13 r1 paper for the iteration scheme on $r=1$ shapes; tracer pairs adapted to multi-corner shapes are open. The technique of refining "$\neq 0$" to an exact $(q+1)$-monomial via $b'=1$ propagation appears robust and may transfer.

Combinatorial reading. The exponent $\binom{k}{2}$ is suggestive: it is the number of inversions in the longest word of S_k. The all-swap chain in Lemma 3 executes a $(k-1)$-cycle of length $k-2$, contributing $k-2$ inversions to the permutation $T'_- \mapsto T_*^{(\lambda)}$. The R'_n trace in Lemma 2 contributes $k-1$ same-row pairs. Summing $(k-2)+(k-1)+\binom{k-1}{2}=\binom{k}{2}$ recovers the answer inductively. A direct combinatorial interpretation of the exponent remains open.